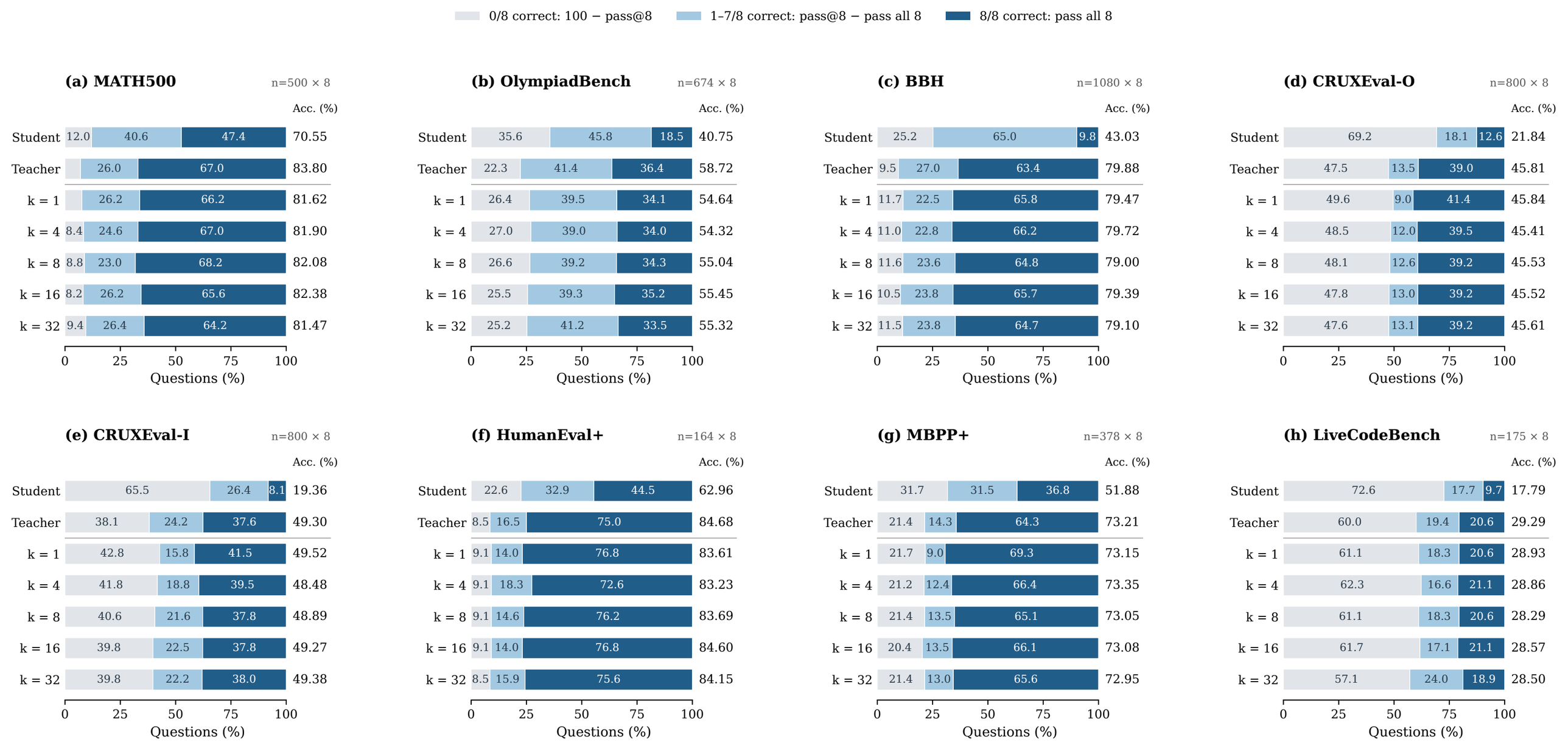
0/8 correct: 100 − pass@8
1–7/8 correct: pass@8 − pass all 8
8/8 correct: pass all 8
(a) MATH500
n=500 × 8
Acc. (%)
Student 12.0 40.6 47.4 70.55
Teacher 26.0 67.0 83.80
k = 1 26.2 66.2 81.62
k = 4 8.4 24.6 67.0 81.90
k = 8 8.8 23.0 68.2 82.08
k = 16 8.2 26.2 65.6 82.38
k = 32 9.4 26.4 64.2 81.47
0 25 50 75 100
Questions (%)
(b) OlympiadBench
n=674 × 8
Acc. (%)
Student 35.6 45.8 18.5 40.75
Teacher 22.3 41.4 36.4 58.72
k = 1 26.4 39.5 34.1 54.64
k = 4 27.0 39.0 34.0 54.32
k = 8 26.6 39.2 34.3 55.04
k = 16 25.5 39.3 35.2 55.45
k = 32 25.2 41.2 33.5 55.32
0 25 50 75 100
Questions (%)
(c) BBH
n=1080 × 8
Acc. (%)
Student 25.2 65.0 9.8 43.03
Teacher 9.5 27.0 63.4 79.88
k = 1 11.7 22.5 65.8 79.47
k = 4 11.0 22.8 66.2 79.72
k = 8 11.6 23.6 64.8 79.00
k = 16 10.5 23.8 65.7 79.39
k = 32 11.5 23.8 64.7 79.10
0 25 50 75 100
Questions (%)
(d) CRUXEval-O
n=800 × 8
Acc. (%)
Student 69.2 18.1 12.6 21.84
Teacher 47.5 13.5 39.0 45.81
k = 1 49.6 9.0 41.4 45.84
k = 4 48.5 12.0 39.5 45.41
k = 8 48.1 12.6 39.2 45.53
k = 16 47.8 13.0 39.2 45.52
k = 32 47.6 13.1 39.2 45.61
0 25 50 75 100
Questions (%)
(e) CRUXEval-I
n=800 × 8
Acc. (%)
Student 65.5 26.4 8.1 19.36
Teacher 38.1 24.2 37.6 49.30
k = 1 42.8 15.8 41.5 49.52
k = 4 41.8 18.8 39.5 48.48
k = 8 40.6 21.6 37.8 48.89
k = 16 39.8 22.5 37.8 49.27
k = 32 39.8 22.2 38.0 49.38
0 25 50 75 100
Questions (%)
(f) HumanEval+
n=164 × 8
Acc. (%)
Student 22.6 32.9 44.5 62.96
Teacher 8.5 16.5 75.0 84.68
k = 1 9.1 14.0 76.8 83.61
k = 4 9.1 18.3 72.6 83.23
k = 8 9.1 14.6 76.2 83.69
k = 16 9.1 14.0 76.8 84.60
k = 32 8.5 15.9 75.6 84.15
0 25 50 75 100
Questions (%)
(g) MBPP+
n=378 × 8
Acc. (%)
Student 31.7 31.5 36.8 51.88
Teacher 21.4 14.3 64.3 73.21
k = 1 21.7 9.0 69.3 73.15
k = 4 21.2 12.4 66.4 73.35
k = 8 21.4 13.5 65.1 73.05
k = 16 20.4 13.5 66.1 73.08
k = 32 21.4 13.0 65.6 72.95
0 25 50 75 100
Questions (%)
(h) LiveCodeBench
n=175 × 8
Acc. (%)
Student 72.6 17.7 9.7 17.79
Teacher 60.0 19.4 20.6 29.29
k = 1 61.1 18.3 20.6 28.93
k = 4 62.3 16.6 21.1 28.86
k = 8 61.1 18.3 20.6 28.29
k = 16 61.7 17.1 21.1 28.57
k = 32 57.1 24.0 18.9 28.50
0 25 50 75 100
Questions (%)